

Fourier Series

This section assumes that you have already watch [3b1b's video on Fourier transformation](#). You should know where this formula comes from and what does it mean:

$$X(\omega) = \int_{\mathbb{R}} x(t)e^{-i\omega t} dt,$$

where $\omega = 2\pi f$ and f is the frequency in Hertz.

Let's start with a **periodic function**, $x(t)$ with a period T . This means that $x(t)$ is completely defined by $t \in [0, T]$, since $x(t + T) = x(t)$. If we wish to see what frequencies are present in this periodic function, we may perform a Fourier transform on it:

$$X(\omega) = \int_{\mathbb{R}} x(t)e^{-i\omega t} dt$$

We can split the integral into intervals of length T and rewrite the integrals so that they all the same bounds:

$$\begin{aligned} X(\omega) &= \dots + \int_{-\frac{T}{2}-T}^{-\frac{T}{2}} x(t)e^{-i\omega t} dt + \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega t} dt + \int_{\frac{T}{2}}^{\frac{T}{2}+T} x(t)e^{-i\omega t} dt + \dots \\ &= \dots + \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t-T)e^{-i\omega(t-T)} dt + \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega t} dt + \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t+T)e^{-i\omega(t+T)} dt + \dots \\ &= \sum_{k=-\infty}^{\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t+kT)e^{-i\omega(t+kT)} dt. \end{aligned}$$

We now take the advantage of the periodicity of $x(t)$, and replace $x(t+kT)$ with $x(t)$:

$$\begin{aligned} X(\omega) &= \sum_{k=-\infty}^{\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega(t+kT)} dt \\ &= \sum_{k=-\infty}^{\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega t} e^{-i\omega kT} dt \\ &= \sum_{k=-\infty}^{\infty} \left(e^{-i\omega kT} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega t} dt \right). \end{aligned}$$

Since the integral does not depend on k , we can take it out of the summation:

$$X(\omega) = \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-i\omega t} dt \right) \left(\sum_{k=-\infty}^{\infty} e^{-i\omega kT} \right).$$

We shall not prove it here, but as in the 3b1b's video, we can assume that the summation

$$\sum_{k=-\infty}^{\infty} e^{-i\omega kT}$$

is going to be zero for most values of ω , except for $\omega_0 = 2\pi \frac{1}{T}$ and its multiples. In other words, the summation (and thus $X(\omega)$) is non-zero only when $\omega = n\omega_0$ for some integer n .

Now we let's consider $X(n\omega_0)$ since all other values of ω are zero:

$$X(n\omega_0) = \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-in\omega_0 t} dt \right) \left(\sum_{k=-\infty}^{\infty} e^{-in\omega_0 kT} \right).$$

Here,

$$\sum_{k=-\infty}^{\infty} e^{-in\omega_0 kT} = \sum_{k=-\infty}^{\infty} e^{-i2\pi n k} = \sum_{k=-\infty}^{\infty} 1 = \infty,$$

the value $X(n\omega_0)$ blows up to infinity. Our attempt to find the Fourier transform of a periodic function has led us to a weird function which is **zero everywhere except for $\omega = n\omega_0$ where it is infinite**. This is not a function in the usual sense, in order to work with it, mathematicians have invented a new kind of functions, from which the most famous one is the Dirac delta function, $\delta(x)$, defined as

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases},$$

with the property that

$$\int_{\mathbb{R}} \delta(x) dx = 1 \text{ and that } \int_{\mathbb{R}} f(x) \delta(x - a) dx = f(a).$$

Intuitively speaking, the Dirac delta function is a function that has a big *spike* at $x = 0$. And our $X(\omega)$ is just a function that has *spikes* at $\omega = n\omega_0$ for all integers n . Hence, we should be able to find a way to express $X(\omega)$ in terms of the Dirac delta function. However, we need to be careful with the *height* of the spikes even though they are infinite. For example,

$$\int_{\mathbb{R}} \delta(x) dx = 1 \quad \int_{\mathbb{R}} 2\delta(x) dx = 2,$$

even though both $\delta(x)$ and $2\delta(x)$ are infinite at $x = 0$. We are going to omit the proof here, but it can be shown that

$$\sum_{k=-\infty}^{\infty} e^{-i\omega kT} = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0).$$

The summation of delta functions on the right-hand side is called the **Dirac comb function** or an **impulse train**. Therefore,

$$\begin{aligned} X(\omega) &= \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-i\omega t} dt \right) \left(\frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0) \right) \\ &= \frac{2\pi}{T} \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-i\omega t} dt \right) \left(\sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \right). \end{aligned}$$

Now, we can reconstruct $x(t)$ from $X(\omega)$ by taking the inverse Fourier transform:

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{\mathbb{R}} X(\omega) e^{i\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2\pi}{T} \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-i\omega t} dt \right) \left(\sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \right) e^{i\omega t} d\omega \end{aligned}$$

Due to the property of the Dirac delta function, the integral over $\mathbb{R}$ becomes a summation of the integrand evaluated at $\omega = k\omega_0$:

$$x(t) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-ik\omega_0 t} dt \right) e^{ik\omega_0 t}.$$

For the sake of convenience, we define

$$c_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-ik\omega_0 t} dt,$$

so that we can write

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t}.$$

This is the Fourier series representation of a periodic function $x(t)$ with period T . The coefficients c_k are known as the Fourier coefficients and they capture the contribution of each frequency component $k\omega_0$ to the overall signal.

Data Sampling Process

Let's say that the true signal is $x(t)$, and we want to sample it at a rate of $f_s = \frac{1}{T_s}$ samples per second. A way to mathematically model the sampling process is to multiply $x(t)$ with an **impulse train** $p(t)$, which is a periodic function with period T_s , defined as

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s).$$

Since we know that it is periodic with period T_s , we can write its Fourier series representation as

$$p(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_s t},$$

where $\omega_s = 2\pi f_s$ is the angular frequency of the sampling process, and c_k are the Fourier coefficients of the impulse train. The Fourier coefficients can be computed as

$$\begin{aligned} c_k &= \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} p(t) e^{-ik\omega_s t} dt \\ &= \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \left(\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right) e^{-ik\omega_s t} dt, \end{aligned}$$

since the integrand is non-zero only when $t = 0$ and $n = 0$, we have

$$c_k = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) e^{-ik\omega_s t} dt = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) dt = \frac{1}{T_s} = f_s.$$

Hence, the impulse train can be expressed as

$$p(t) = \sum_{k=-\infty}^{\infty} f_s e^{ik\omega_s t} = f_s \sum_{k=-\infty}^{\infty} e^{ik\omega_s t}.$$

Now as we want to analyze the frequency content of the sampled signal, we can take the Fourier transform of the sampled signal $x_{s(t)} = x(t)p(t)$:

$$\begin{aligned}
 \mathcal{F}\{x_{s(t)}\} &= \mathcal{F}\{x(t)p(t)\} \\
 X_s(\omega) &= \int_{\mathbb{R}} x(t)p(t)e^{-i\omega t} dt \\
 &= \int_{\mathbb{R}} x(t) \left(\sum_{k=-\infty}^{\infty} f_s e^{ik\omega_s t} \right) e^{-i\omega t} dt \\
 &= \int_{\mathbb{R}} \left(\sum_{k=-\infty}^{\infty} f_s x(t) e^{-i(\omega - k\omega_s)t} \right) dt \\
 &= f_s \sum_{k=-\infty}^{\infty} \int_{\mathbb{R}} x(t) e^{-i(\omega - k\omega_s)t} dt.
 \end{aligned}$$

Notice that the integral in the last line is just the Fourier transform of $x(t)$ evaluated at $\omega - k\omega_s$, so we can write

$$X_s(\omega) = f_s \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s). \blacksquare$$